

VAMSHI ORUGANTI

Mechatronics Engineer

Experience 5.6+ Years

Vamshi Oruganti - Embedded Engineer - Project Contributions

COMPLETED PROJECTS

- CPAP/BIPAP/HFNC Devices developed and software functionality designed and developed.
- Engineering Model CPAP/Auto CPAP device developed, as per standard ISO 80601-2-70.
- Developed CPAP, Timed, Spon, S/T, AVAPS and HFNC algorithms /logics.
- Display GUI designed and logics implemented in controller as well in UART based screen.
- Therapy data storage in a EEPROM / SD Card(Efficient logics implemented).
- Blowers, Pressure sensor, Differential Pressure sensor(Flow Sensor), SD Card, EEPROM, RTC and Topway display/Dwin Display and Wizfi360 / RNWF11/ RNWF02.
- As a mechatronics engineer integration issues solved, coordinated with Hardware, mechanical, UI/UX and Software team for mobile phone application development.

Skills

- PIC18F/PIC24F/PIC32 controller
- MPLAB X IDE
- Debugging skills
- PWM, Timers, ADC and EXT Interrupt Pins
- UART, I2C, SPI, SDHC and QSPI
- Boot Loader and OTA
- Interrupt and State Machine coding
- Bare metal programming and C
- Display Design and logic implementation
- Closed loop(PID and Fuzzy)
- Actuators and sensors
- Sensor calibration techniques
- Smart Hardware Circuits Logics
 - Power fault condition
 - Noise Elimination at ADC
 - Voltage Dividers, Pull up and Pull down

Vamshi Oruganti - Sr. Engineer - Project Contributions

COMPLETED PROJECTS

P1. Electro-Polishing

- Design and developed a relay based circuit with safety interlocks and operator friendly to monitor and control operations
- Integration with existing Battery Charger
- BOM preparation and components selections

P2. Ampoule Making Machine

- Integration of HMI and PLC and configuring the pneumatic, electrical components as per circuit
- Fine tuning the machine to produce different sizes of ampoules
- Mechanical assembly
- Sensors Integration and troubleshooting

P3. RH Monitoring & alarm(IIoT)

- Design and developed a RH monitor system with cloud based data monitoring with user access
- RH monitor also includes alarm which gives signal when the RH measured is out of set value
- BOM and Components selection
- Program development, RH sensor integration.

P4. Electrode Shearing Online

- Design and developed a control system to auto activate the cylinder to shear the electrode to required width that can integrate with existing coating machine
- BOM, Components selection
- Electrical calculations i.e encoders, pulse counter

P5. Tank Level Control

- Design and developed a HMI and PLC based system that can maintain the solution in the line tanks to required level by circulating solution from Main central tank
- BOM, Component selection

P6. Filling Machine

- Design and developed a HMI and PLC based system that can pump the electrolyte to achieve the set point i.e Milliliters.
- Stepper motor with ceramic pump.
- Load cell integration to ensure the set point.

Vamshi Oruganti - Sr. Engineer - Project Contributions

COMPLETED PROJECTS

P7. Weight Sorting

- Design and development of HMI and PLC based system that can sort the plates according to the weight
- BOM preparation, Components selection
- Mechanical design and development
- Servo motor configuration and integration to PLC

P8. Semi Automation Projects

- Design and development of Pneumatic based circuits
- BOM, Component selection
- Mechanical design and development

Softwares

- PLC - Gx Works2 / Gx Works3
- HMI - GT Designer 3
- Servo - MR Configurator2
- Solidworks
- VS Code
- Pycharm and Thony
- Festo Pneumatics
- Festo Hydraulics
- Arduino IDE and Cloud Integration
- CodeSys Basics

Skills

- Start to end project coordination with the end user, design and develop a program for PLC and HMI
- Sensor selection and integration of PLC with other devices.
- Solid knowledge in PLC(AI/AO/DI/DO/PWM)
- PLC(Configuring, Testing, and Troubleshooting)
- Motion Controls(DC, Servo, Stepper Motors)
- Develop the control systems
- Mechanical Motion Calculations
- Electrical BOM and cost estimation
- Pneumatic Circuits and Component selection
- 3D Printer - FDM
- Vendor / Supplier Identification
- Microcontrollers(Esp32 S2,S3 and STM32(Portenta H7))
- Microprocessors(Raspberry pi and Jetson Nano)
- Microcontroller communication(I2C, SPI and CAN)
- Proportional valves and selection
- Interdisciplinary skills and Interdisciplinary collaboration

Vamshi Oruganti - Sr. Engineer - Project Contributions

WORKED ON PROJECTS

P1. Set Point LPM

- Design and developed a PID circuit, Programming in PID module
- Closed loop circuit, BOM, Component selection

P2. Retrofit - Star Robot

- Supplier identification and validation
- Middle man between end user and supplier
- Technical clarification
- Robot programming

P3. Dual Spot Weld Machine

- Supplier identification and validation
- Middle man between end user and supplier
- Technical clarification

P4. Glueing Robot

- Supplier identification and validation
- Middle man between end user and supplier
- Technical clarification

P5. Ampoule Filling

- Design and developed of program for find out the center point of the ampule neck hole i.e $0.5\text{mm} \pm 0.1\text{mm}$.
- Design and developing motion control system
- BOM, Components selection
- Mechanical design and development

- Computer vision applications
- Write a program in different controllers
- Integrate the mechanical and electronics
- Develop the electronic and electrical circuits
- Robotic arm program

- > Design and develop a Mechatronics based system
- > IoT applications
- > Assembly the mechanical part
- > Able to design the mechanical parts
- > Sensors and Actuators Selection

Vamshi Oruganti - Graduate Engineer Trainee - Project Contributions

COMPLETED PROJECTS

- Designed a Li-ion battery pack and made an E-bike prototype using a BLDC hub motor.
- Developed Embedded software for the battery packs to monitor and regulate the charging and discharge of batteries. The battery characteristics monitored are over-voltage protection, under-voltage protection, current protection, reverse polarity protection temperature, capacity, state of charge, etc. Also designed a BMS circuit with over-voltage protection, under-voltage protection and reverse polarity protection. Gain knowledge of the Electrical Vehicle functionality and Concepts.
- Worked on communication protocols(I2C , SPI and CAN)

Skills

- ESP32
- Embedded Programming
- VS Code
- Li-ion Battery Pack
- Li-ion cell Selection
- BLDC Motors
- BLDC Controllers
- E-Bike & Battery testing
- Mechanical & Electrical assembly
- Microprocessors(Raspberry pi)
- Circuit design
- Git

Vamshi Oruganti - Technical Assistant - Project Contributions

COMPLETED PROJECT

- Designed and fabricated an Automatic TIG Welding system using Mechatronics concepts.
- Skillfully worked with various sensors, DC gear motors, Servo motors, drivers, encoders, and communication protocols (I2C, SPI).
- Developed the necessary microcontroller code and an intuitive display interface for the system.
- Maintained a primary focus on ensuring constant linear momentum during the TIG welding operation.
- This project was sponsored by JNTUH (Jawaharlal Nehru Technological University Hyderabad) under the reference JNTUH/TEQIP-III/CRS/2019/MECH/06.

Skills

- Motion Controls
- DC, Servo, Stepper Motors & drives
- BOM and Components selection
- Encoder calculations
- Closed loop Control
- Sensors and actuators
- Embedded C
- Problem solving techniques

COMPLETED PROJECTS

- Agriculture Robot
- Robo Boat
- Advance Multipurpose Robo Boat
- AGV-Fuzzy Logic
- IoT-Bus Tracking
- Automatic Gear Shifting - Bikes
- Pneumatic Projects
- Hydraulic Projects

Skills

- Mechatronics and Control Systems
- Pneumatics, Hydraulics
- E- Pneumatics, E-Hydraulics.
- Sensor Technology
- Internet Of Things (IoT)
- Python, C-Language
- PLC (Ladder logic)
- Pneumatic Circuits
- Hydraulic Circuits
- Solidwork and AutoCAD Basics
- Matlab-Basics

Thank You